

RF Sim for KiCad – Verification Report

Release 0.26.4: simulation compared with closed-form references, with TI's measurements of the CC-Antenna-DK2 kit and with own VNA measurements

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Summary

RF Sim is a KiCad 10 plugin that builds a 3D model from the open board and simulates printed antennas and RF matching networks with the open-source FDTD solver openEMS. This report states how far its results can be trusted in release 0.26.4. It uses three references of rising difficulty.

- **Closed-form references (14 generated boards).** Lines, stubs and single parts agree with theory within the tolerances. Open points: with parts inside the field solver instead of ports with circuit math, the cutoff of a C-L-C low-pass moves by −4.6 % and a pi network at 2.4 GHz differs by 0.75 dB to 1.06 dB. The mesh of a 2.4 GHz inverted-F board is not converged.
- **TI's measurements (8 antenna boards).** Resonance frequencies agree within 0 % to 4 % below 1 GHz and 3 % to 9 % at 2.4 GHz. Total efficiency agrees within ±3 dB in 17 of 19 values. The feed resistance is too low, so the simulated bands are narrower. The directivity is 1.8 dB to 3.5 dB below TI's values on six of the eight boards.
- **Own measurements (6 boards, NanoVNA-F V2).** Board #9 agrees well: its four SWR minima lie within −1 % to 3 % of the measurement. On #2, #6 and #10 the resonance differs by 5 % to 8 %. The measurement of #1 is most likely invalid (a setup problem on this board), and on #4 neither the simulation nor TI's data match it. The own setup is simple (section 4), so these large deviations may come from the measurement rather than the simulation.
- **Warning – simulation time.** openEMS runs on the CPU only. On the test PC (Intel Core i5-4590T, 2014) one antenna board needs 10 min to 3.6 h, one matching network 14 min to 2.5 h. Work on shorter run times is in progress (section 5); it is not part of release 0.26.4.

The plugin is a research project to see what an open-source solver can do inside KiCad. It is not a finished product. Much more validation is needed before its values can be fully trusted. Results are a first estimate: check important designs with a measurement or a professional tool.

1 What was verified

Reference	What is compared	Boards
Closed-form theory	line impedance, ε_{eff}, stub notch, single parts, dipole, patch; parts inside the field solver against ports with circuit math	14 generated test boards
TI SWRA496A	SWR, feed impedance, total efficiency, directivity of the kit's antennas as built	CC-Antenna-DK2 #1, 2, 4, 6, 9, 10, 11, 12
Own VNA measurement	S11 at the coax connector as built	#1, 2, 4, 6, 9, 10 (#11: connector fault)

Plugin version. All results are those of release 0.26.4 with openEMS 0.37.0. The antenna simulations of the TI boards were run on 9–10 November. For each of the eight boards the model file was generated again with the code of release 0.26.4: it is byte-identical (same SHA-256) to the model that was simulated. The values of the generated test boards are those of the bench run published with release 0.26.4.

Files. The test boards are published for your own validation in `kicad-rfsim-boards-<version>.zip` (`https://github.com/at-kat-454/kicad-rfsim-releases`): folder `validation` for the generated boards, folder `ti_dk2` for the TI boards. They contain the KiCad projects, the setups and the reference values, but no simulation results: results of different plugin versions are not consistent with each other, so each user simulates the boards with the version in use.

2 Closed-form references

Every case runs through the full plugin chain: board import, setup, mesh, openEMS and circuit math.

Case	Simulation vs. reference	Within tolerance
50 Ω microstrip, 0.2 mm FR-4	Z_0 51.6 Ω vs. 50.0 Ω; ε_{eff} 3.28 vs. 3.33	yes
Grounded coplanar line, 0.25 mm track	Z_0 60.6 Ω vs. 61.0 Ω; ε_{eff} 3.20 vs. 3.08	yes
Grounded coplanar line, 1.6 mm FR-4	Z_0 48.4 Ω vs. 50.0 Ω; ε_{eff} 2.85 vs. 2.78	yes
Open $\lambda/4$ stub, 2 GHz	notch 2.046 GHz (+2.3 %)	yes
Series C, series L, shunt C, shunt L (0402)	$\lvert S_{21}\rvert$ within 0.02 dB to 0.72 dB of the ideal circuit at 868 MHz and 2 GHz; parts inside the solver vs. ports within 0.32 dB	yes
C-L-C low-pass, $f_c \approx 1$ GHz	cutoff 927 MHz (ports) vs. 884 MHz (parts inside the solver): −4.6 %, limit 2 %	no
Pi network 1 pF / 2.7 nH / 1 pF, 2.4 GHz	parts inside the solver vs. ports 0.75 dB to 1.06 dB, limit 0.5 dB	no
Half-wave strip dipole	resonance at 0.469 λ, 71.8 Ω, 2.07 dBi, 98 %	yes
Dipole in an ABS housing	resonance −0.5 % vs. free space	yes
Patch, 2.45 GHz, 1.6 mm FR-4	2.377 GHz (−3.0 %), 6.0 dBi	yes
2.4 GHz board with inverted-F antenna	resonance 2.366 GHz (−3.0 %), efficiency 89 %; Fast → Automatic −3.7 %, Automatic → Accurate +0.57 %: mesh not converged	partly

The low-pass and the pi network both concern a matching network on a thick two-layer board. In release 0.26.4 every tuned part gets a port on each pad down to the ground plane. These port columns through the board add their own inductance (0.6 nH to 1 nH on 1.5 mm FR-4). Section 5 describes the change under test.

3 TI CC-Antenna-DK2 boards

3.1 Model

The KiCad boards were converted automatically from TI's Gerber files, drill file and bill of materials by a script (`build_ti.py`); they were not drawn by hand. The copper is imported as polygons, not as tracks and zones. The connector pads are rectangles around TI's paste openings, so their exact shape is approximated. The board is FR-4, 1.47 mm thick, with ε_r 4.2 (TI's value). The matching parts are lumped elements with the values of the kit (inductor Q 30 at the band centre). The feed is a 50 Ω port at the signal pad of the coaxial connector (Murata MM5829), to the ground beside it. The mesh is the plugin's automatic mesh. Nothing was tuned to a measurement. The simulated quantity is S11 at the coax pad, as built: the same plane as the measurements.

3.2 Own measurement setup

- NanoVNA-F V2, a thin coaxial pigtail from the board's coaxial connector to an SMA bulkhead.
- Calibration at the coax plane with the kit's calibration boards #13 (short), #14 (load) and #15 (open). Check: the load reads −58.6 dB at 900 MHz after calibration (figure 1, left).
- Sweep: 0.7 GHz to 2.7 GHz in 20 MHz steps (101 points) for the 868 MHz and 2.4 GHz boards; 400 MHz to 550 MHz in 1.5 MHz steps for #4 and #10.
- The board hangs on the cable or is held at its edge by a plastic clamp. The cable is not choked.
- Board #11 was not measured: its connector gave a constant open, most likely a contact problem of the mating plug. The file saved for #12 holds the data of #10, so #12 is left out as well.



Figure 1: Top, from left: calibration check on the load board #14; measurement of #9 and #2 with the board in a clamp. Bottom: the boards of the kit.

3.3 Resonance and match

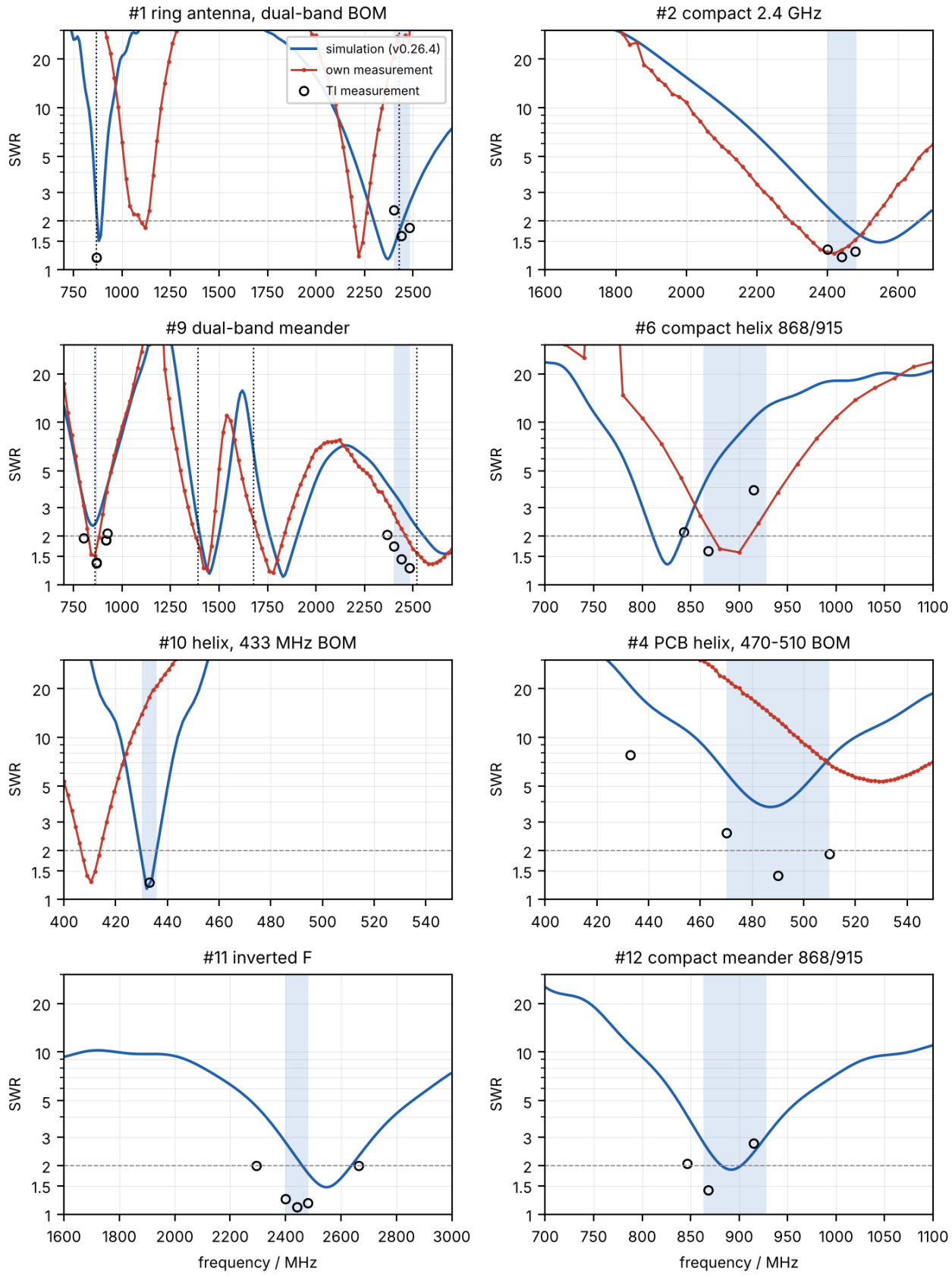


Figure 2: SWR at the coax connector, as built. Blue: simulation. Red: own measurement. Circles: TI's measured values, dotted lines: TI's SWR minima. Shaded: design band. #11 and #12 have no own measurement.

Board	Simulation: SWR minimum	Own measurement	TI
#1 low band	880 MHz (SWR 1.51)	*(likely invalid)* no resonance near 868 MHz; minimum at 1 120 MHz	866 MHz
#1 high band	2 370 MHz (1.16)	*(likely invalid)* 2 220 MHz (1.22)	2 430 MHz
#2	2 552 MHz (1.48)	2 420 MHz (1.26)	SWR 2 band 2331–2590 MHz
#4	487 MHz (3.71)	529 MHz (5.34)	SWR 1.40 at 490 MHz
#6	826 MHz (1.33)	900 MHz (1.58)	SWR 1.62 at 868 MHz
#9 (4 minima)	850 / 1450 / 1830 / 2670 MHz	860 / 1440 / 1780 / 2600 MHz	860 / 1393 / 1678 / 2520 MHz
#10	432 MHz (1.16)	410 MHz (1.28)	SWR 1.27 at 433 MHz
#11	2 545 MHz (1.47)	not measured (connector)	SWR 2 band 2295–2663 MHz
#12	892 MHz (1.88)	–	SWR 1.42 at 868 MHz

- **Best case #9.** The four minima of the simulation lie within -1.2, $+0.7$, $+2.8$ and $+2.7$ % of the own measurement. At 868 MHz the reflection coefficients differ by $|\Delta\Gamma| = 0.19$.
- **#2, #6, #10.** The simulated resonance is shifted: #2 +5.5 %, #6 −8.2 %, #10 +5.4 % against the own measurement. For #10 the simulation agrees with TI (433 MHz), the own measurement does not (410 MHz).
- **#2: own measurement equals TI.** At 2 400 MHz to 2 480 MHz the own measurement and TI differ by $|\Delta\Gamma| \leq 0.14$, the simulation by 0.22 to 0.32. This is the clearest error of the simulation on a 2.4 GHz board: the band lies about 100 MHz too high.
- **#1: measurement most likely invalid.** The measured SWR minima (1120 and 2 220 MHz) lie far from both TI's (866 / 2 430 MHz) and the simulation's (880 / 2 370 MHz). I attribute this to a setup problem with this particular board rather than to the board itself. #1 is a 26.5 mm round board: at 868 MHz the cable is part of the radiator (in the earlier TI report, a 100 mm cable in the model raised the efficiency at 868 MHz from 0.02 % to 36 %).
- **#4.** Neither the simulation nor the own measurement show TI's match at 490 MHz (TI: SWR 1.40; simulation: 3.71 at 487 MHz; measurement: 5.34 at 529 MHz).

3.4 Efficiency and directivity (against TI)

Board	Frequencies	Total efficiency, simulation	TI
#2	2400 / 2440 / 2480 MHz	67 / 73 / 77 %	64 / 62 / 71 %
#11	2400 / 2440 / 2480 MHz	66 / 73 / 79 %	71 / 69 / 77 %
#9	868 / 915 / 2440 MHz	77 / 60 / 69 %	80 / 43 / 67 %
#6	868 / 915 MHz	49 / 24 %	63 / 29 %
#12	868 / 915 MHz	42 / 52 %	46 / 31 %
#4	470 / 490 / 510 MHz	22 / 35 / 29 %	15 / 37 / 48 %
#10	433 MHz	17 %	26 %
#1	868 / 2440 MHz	0.02 / 1.3 %	52 / 21 %

- 17 of 19 values lie within ±3 dB of TI (largest: #12 at 915 MHz +2.3 dB, #4 at 510 MHz −2.1 dB). Only board #1 fails; as a bare board it cannot reach TI's value (Chu limit, see the earlier TI report). 14 of 19 lie within 15 percentage points.
- Directivity: TI measured 3.7 dBi to 5.9 dBi on every board. The simulation gives 4.6 dBi to 4.9 dBi on #9 at 2.44 GHz and on #11, and 3.5 dBi to 5.2 dBi on #1, but only 1.7 dBi to 2.5 dBi on the other boards. TI's chamber includes cable and positioner, and its peak value is raised by the scatter of its samples (detailed in the earlier TI report).

4 Discussion of the deviations

I am a student, and this comparison has clear limits. This section lists them on both sides, so that the numbers above can be weighed accordingly. Three sources are separated: my own measurement, TI's reference data and the simulation model.

4.1 Own measurement

The setup was chosen for what was available, not for antenna metrology. Its main weaknesses:

- **Common-mode current on the cable.** The coaxial pigtail is not choked (no ferrites, no sleeve balun) and leaves the board at an arbitrary angle. Most of the kit's boards have a ground plane shorter than $\lambda/4$ at the lower band. Then the outer conductor of the cable carries antenna current and becomes part of the radiator. Resonance frequency and feed resistance then depend on the cable length and route, which I did not control or record.

- **Calibration standards.** The open, short and load were the kit's calibration boards #13–#15. I had no characterised definitions for them (open capacitance, short inductance, offset delay), so they were treated as ideal. The resulting phase error grows with frequency and is unknown. The load check (−58.6 dB at 900 MHz) confirms the load, not the reactive standards. On #2, $|\Gamma| > 1$ between 1.2 and 1.42 GHz shows that the calibration was not valid everywhere.
- **Instrument and sweep.** A NanoVNA has a far lower directivity and dynamic range than a laboratory VNA. The sweeps from 0.7 to 2.7 GHz used 101 points (20 MHz steps): a minimum is resolved to about ± 1 % at 868 MHz, and a narrow resonance can fall between two points.
- **Near field.** The plastic clamp at the board edge and the table were within a fraction of a wavelength of the antennas. Their dielectric loading lowers the resonance by an amount I did not quantify.
- **Repeatability.** The measurements were not repeated with a re-routed cable, so the spread of the setup is unknown. It may well exceed the simulation error on the small boards.

For these reasons I regard the large deviations of the own measurement on #1, #4 and #10 as most likely caused by the setup. The agreement on #2 (own measurement within $|\Delta\Gamma| \leq 0.14$ of TI) and on #9 (four minima within 3 % of the simulation) shows that the setup gives usable results where the board is large enough or the cable effect is small.

4.2 TI's reference data

TI measured in an anechoic chamber with a cable and a positioner arm in place. The earlier TI report showed that TI's pattern values at the poles vary by up to 11 dB with the rotation, which a bare board cannot do. TI's data therefore also contain cable effects, and they are a reference for the board *in TI's fixture*. The SWR values were read from TI's plots and markers, with an uncertainty of about ± 15 MHz for the minima.

4.3 Simulation model

- **Bare board.** The model contains no cable and no fixture. This is the largest known difference to both measurements, and it explains the low feed resistance only in part: it is about half of TI's at 2.4 GHz, where the boards are not small against the wavelength.
- **Board geometry.** The boards were converted automatically from Gerber files. Copper outlines are polygons on a stepped FDTD grid, so curved structures (the ring of #1, meander corners) are staircased. Connector pads are approximated. Solder mask is not modelled; it would lower the resonance by an estimated 1–2 %.
- **Material.** FR-4 with $\varepsilon_r = 4.2$ and $\tan\delta = 0.02$, constant over frequency, copper without surface roughness. The real ε_r of FR-4 falls with frequency, but its value at 2.4 GHz on these boards is not known.
- **Feed.** A lumped 50 Ω port at the connector pad, to the ground beside it. The connector body and its transition are not in the model.
- **Parts.** Lumped elements with a fixed Q at the band centre; no self-resonance, no package or pad parasitics beyond the copper geometry.
- **Mesh.** The automatic mesh was used. Mesh convergence was checked only on single boards (#11: Fast → Automatic moves the resonance by 0.4 %); on the others the mesh error is not quantified.

4.4 What can be concluded

- Where the measurement setup is least critical (#2 against TI, #9 against the own measurement), the simulated resonances agree within about 1–5 %, and the total efficiency within ± 3 dB on all boards but #1.
- The systematic underestimation of the feed resistance and of the bandwidth is real for the bare-board model. Whether it is a modelling error or the missing cable cannot be separated with the present data.
- A meaningful validation of the small sub-GHz boards needs a choked cable or a measurement with the cable included in the model, repeated measurements, and characterised calibration standards.

5 Simulation time – a warning

The simulation is slow. openEMS solves the fields on a regular grid in the time domain, on the CPU only. It has no GPU engine. A fine copper detail forces small cells everywhere along its mesh lines, and the smallest cell sets the time step. Commercial tools avoid this with conformal cells, adaptive meshes and GPU solvers.

Board	Setup	Cells	Runs	Timesteps per run	Time (sum of runs)
#1	antenna, Fast	0.41 M	1	22 260	10 min
#1	RF chain, Fast	0.14 M	11	12 000	14 min
#2	antenna	0.51 M	1	19 170	7 min
#2	RF chain	0.16 M	7	21 168	20 min
#6	antenna, Automatic	1.30 M	1	63 492	47 min
#6	RF chain, Automatic	0.38 M	7	56 000–59 000	63 min
#9	antenna, Automatic	1.76 M	1	60 888	58 min
#9	RF chain, Automatic	0.48 M	11	55 000–60 000	136 min
#4	antenna, Automatic	2.74 M	1	160 000	216 min
#4	RF chain, Automatic	0.39 M	9	106 000–115 000	152 min

Test PC: Intel Core i5-4590T (4 cores, 2.0 GHz, 2014), 30–40 million cell updates per second. The 470 MHz to 510 MHz board #4 needs most time: the antenna is small against the wavelength and rings long, and the cells stay fine.

Work in progress (not in release 0.26.4). The development version takes over two ideas of the commercial tools. (1) One port across each tuned part instead of one per pad, as the discrete ports of a field solver coupled to a circuit simulator. It needs fewer runs (#6: 5 instead of 7) and removes the port columns through the board. (2) A short broadband excitation pulse. On board #6 the RF chain took 28 s instead of 139 s on the same mesh in the test environment. In the same version the pi network and the low-pass cutoff of section 2 agree within tolerance; a shunt capacitor at 2 GHz does not yet. These changes are being verified and will be part of a later release.

6 Conclusion

- Transmission lines, stubs, single parts and simple antennas agree with closed-form references.
- On real antenna boards the plugin finds the resonances within a few percent and the total efficiency within ± 3 dB on all boards but the smallest. It describes the bare board, not the board with its measurement cable.
- The feed resistance and so the bandwidth are underestimated. Matching values from the plugin are a starting point for the lab, not a replacement for a measurement.
- Simulation times of minutes to hours on an older desktop PC are the main practical limit. Reducing them is the current focus.